Stack Implementation Using Linked List

C Program Code and Execution Output

PROGRAM SOURCE CODE

```
#include<stdio.h>
#include<stdlib.h>

struct node
{
    int data;
    struct node *next;
};

struct node *top = NULL, *temp, *newNode;

void push(int data)
{
    newNode = (struct node*)malloc(sizeof(struct node));
    newNode->data = data;
    newNode->next = NULL;
    if(top == NULL)
    {
        top = newNode;
    }
    else
    {
        newNode->next = top;
        top = newNode;
    }
    printf("element pushed:%d\n", data);
}

int pop()
{
    if(top == NULL)
    {
        printf("\nEMPTY STACK");
        return -1;
    }
    else
    {
        temp = top;
        int temp_data = top->data;
        top = top->next;
        free(temp);
        return temp_data;
    }
}

void display()
{
    if(top == NULL)
    {
        printf("\nStack Underflow\n");
    }
    else
    {
```

PROGRAM OUTPUT

```
element pushed:10
element pushed:20
element pushed:30

The stack is:
30--->20--->10--->NULL

element poped 30
element poped 20
The stack is:
10--->NULL
```